

An exact exponential fixed point and metastable shape manifolds in a renormalization cascade of finite-time Lyapunov distributions:

a two-class split of chaotic dynamics

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Abstract

We study a renormalization cascade T on distributions of finite-time Lyapunov exponents (FTLEs): at each level, samples are replaced by mean-normalized absolute differences of independent pairs, and a maximum-likelihood tail exponent is read off. The operator first turned up in our own large-scale computational study of double-pendulum chaos, where its limit $\alpha^* \approx 4$ was conjectured to be a new universal constant. We give it an exact theory here. The exponential distribution turns out to be the *unique* exact fixed point of T (Puri–Rubin), with a closed-form reading $\alpha_{\text{Exp}}^*(f) = 1 + [\frac{1}{f}E_1(\ln \frac{1}{f})]^{-1} = 3.3494$ at $f = 0.2$, confirmed to 0.02%–0.3%, and *every* Gaussian bulk collapses in a single step onto one universal half-normal shape. Once T is linearized exactly, the symmetrization step is seen to halve excess kurtosis at each level, which *derives* the previously empirical “universal” decay constant $r \approx 1/2$; underneath sits a gapless, non-normal spectrum ($\mu(s) = 2/(2-s)$ over a unit-diagonal polynomial tower) whose measured consequence is algebraic relaxation and metastable plateaus. So the celebrated “ $\alpha^* \approx 4$ ” is a plateau reading of a marginal flow rather than an asymptotic constant, and the operator is provably not a tail-index estimator. What survives this reckoning is, if anything, sharper: a *two-class split of chaotic dynamics*. Across eleven FTLE ensembles from five systems, mixed-phase dynamics yields broad, *platykurtic* FTLE bulks that read strictly *above* the exponential fixed point, while fully chaotic, dissipative and near-regular dynamics yield *leptokurtic* bulks that read *below* it. The separation is sign-perfect (11/11; sign test $p = 0.001$; Cohen’s $d = 3.3$ on a same-map/same-protocol control) with statistically zero within-class heterogeneity. An out-of-sample Hénon–Heiles sweep, predicted in advance, flips its matched-null excess exactly where phase space turns from a thin chaotic web to a mixed sea. Read against an exactly known reference point, the shape of the FTLE distribution turns out to be a cheap and sensitive probe of a system’s ergodic architecture.

Keywords: finite-time Lyapunov exponents; chaotic Hamiltonian dynamics; mixed phase space; renormalization fixed point; Chirikov standard map; double pendulum; extreme-value theory; universality class; metastable plateaus

Contents

1 Introduction

2

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2	The cascade operator and its exact structure	4
2.1	Definitions	4
2.2	Scale invariance	5
2.3	The exponential fixed point and its uniqueness	5
2.4	Closed-form readings	5
2.5	Universal Gaussian collapse	6
2.6	Tail conservation — and why it is not realized	6
2.7	The linearized cascade: exact cumulant flow and a gapless, non-normal spectrum	7
3	Systems, data and statistical protocol	8
4	Calibration on synthetic ensembles	9
5	The two-class split of chaotic dynamics	9
6	The metastable origin of “$\alpha^* \approx 4$”	11
7	What the pendulum constants are: frozen ensemble statistics	12
8	Out-of-sample test: Hénon–Heiles across its transition	14
9	Dynamical mechanism, interpretation, and predictions	15
10	A number-theoretic audit of the constants	18
11	Conclusions	18
A	Derivations of the closed-form readings	19
B	Data provenance and reproducibility	20
C	Full synthetic calibration table	20
D	Estimator and extrapolation pathologies	20

1 Introduction

The distribution of finite-time Lyapunov exponents (FTLEs) is one of the richest observables a chaotic system offers. Its mean gives the leading Lyapunov exponent [1, 2], while its width and shape carry the fluctuations of local expansion rates and, via large-deviation theory, the multifractality of the invariant measure [3, 4, 5, 6]. In Hamiltonian systems with a mixed phase space, trajectories that get intermittently trapped near regular islands have their finite-time exponents depressed, which produces long-lived non-Gaussian structure [7, 8]. The associated algebraic decay of Poincaré recurrences is itself conjectured to be universal [9, 10].

This paper is about a specific, deceptively simple *renormalization cascade* on FTLE distributions, one that emerged from our own large-scale computational study of double-pendulum chaos [11, 12] (hereafter, the *source program*). Starting from the chaotic subset of an FTLE sample, the operator repeatedly (i) fits a maximum-likelihood power-law exponent to the upper tail, and (ii) coarse-grains the sample by replacing it with mean-normalized absolute differences of independent pairs. The per-level exponents α_k were observed to converge geometrically to a limit $\alpha^* \approx 4$, robust across hundreds of double-pendulum parameter combinations. Together with a companion level-zero exponent $\alpha_0 \approx 8.3$, this value was conjectured to be a new universal constant of chaotic Hamiltonian dynamics. We subsequently re-measured the whole claim

family ourselves at $n = 5 \times 10^5$ – 5×10^6 per system, and the picture sharpened in two directions. First, the α^* landscape over double-pendulum parameter space turned out to be statistically *flat*: no ridge structure survives a noise null. Second, a class of dynamically distinct systems, namely the dissipative Lorenz-63 attractor, the fully chaotic standard map at $K = 5$, and a near-regular coupled quartic oscillator, clusters at a sharply *different* fixed point, $\alpha^* \approx 3.0$ – 3.3 , while mixed-phase systems (the double pendulum family and the standard map just above the connectedness threshold, $K = 1$) cluster near 3.9 – 4.4 [13].

Two questions therefore demanded an answer. *What does this operator actually measure?* And *is the two-class structure a property of the dynamics or of the statistics?* We answer both.

Contributions.

1. **Exact fixed point (Theorem 1).** The exponential distribution is an exact fixed point of the cascade map, including its mean normalization. By the characterization theorem of Puri and Rubin [14], it is in fact the *unique* distribution satisfying $|X - Y| \stackrel{d}{=} X$. The folding identity behind this is elementary and, as far as we know, had not been connected to Lyapunov statistics before.
2. **Closed-form readings (Propositions 1, 2).** We derive in closed form the maximum-likelihood tail exponent that the pipeline assigns to the exponential fixed point, $\alpha_{\text{Exp}}^*(f) = 1 + [\frac{1}{f}E_1(\ln \frac{1}{f})]^{-1}$ ($= 3.3494$ at $f = 0.2$), confirmed numerically to 0.02%. Companion formulas follow for the half-normal (4.4448), the uniform (9.6417), and a first-order level-zero law for Gaussian bulks, $\alpha_0 \approx 1 + (\text{cv}^{-1} + z_f)/m_f$, each likewise confirmed.
3. **Universal Gaussian collapse (Theorem 2).** Because the location parameter cancels in pair differences, *every* Gaussian bulk lands, after just one cascade step, on the same half-normal shape. All Gaussian-seeded flows therefore coincide from level one onward (measured: $\alpha_1 = 4.448$ – 4.455 across coefficients of variation 0.05–0.33), tracing out a single slow relaxation manifold toward the exponential fixed point.
4. **Metastability (§6).** In deep cascades ($n = 1.6 \times 10^7$, 20 levels), the Gaussian-seeded flow’s Kolmogorov distance to the exponential collapses from 0.34 to 0.03 within five levels and then *stalls* at 0.02–0.03 for at least eight more, while the fitted exponent drifts slowly (≈ -0.04 per level) through precisely the range in which the empirical “constants” were reported. The n -dependence of the double-pendulum plateau ($4.64 \rightarrow 4.12 \rightarrow 3.88$ for $n = 5 \times 10^4 \rightarrow 5 \times 10^5 \rightarrow 5 \times 10^6$) rides this same drift. So “ $\alpha^* \approx 4$ ” turns out to be a metastable plateau reading, not an asymptotic constant.
5. **The operator is not a tail estimator (§4).** Genuine Pareto tails are asymptotically conserved by the cascade (Proposition 3), yet they are *not* recovered by the finite-window fit: pure Pareto inputs with survival indices 2.5/4/6 read 2.4/2.7/2.9 after one step. Reading α^* as a physical tail index, then, is untenable.
6. **The two-class split is real, and its mechanism is identified (§5).** Across eleven ensembles from five systems, mixed-phase dynamics produces broad, near-symmetric, *platykurtic* FTLE bulks (excess kurtosis -1.0 to -1.3) that read strictly above the exponential fixed point, while non-mixed dynamics produces narrow or skewed *leptokurtic* bulks (excess kurtosis $+1.3$ to $+43$) that read below it. A per-system matched-Gaussian null (a folded Gaussian sharing the system’s own mean and variance) gives a sign-perfect separation (11/11 ensembles, $p = 0.001$; 8/8 systems, $p = 0.008$), and the contrast survives the sharpest available control: the same map, integrator, horizon and n at $K = 1$ (mixed) versus $K = 5$ (fully chaotic). A random-effects meta-analysis finds within-class heterogeneity indistinguishable from zero (τ^2 CI $[0, 0.012]$, Cochran $p = 0.36$), so the split looks like a *discrete class boundary*, not a soft gradient.

7. **Exact linearization; the decay constant $r \approx 1/2$ derived (§2.7).** The symmetrization step $A - B$ annihilates all odd cumulants and rescales the standardized cumulant of order $2m$ by exactly 2^{1-m} : excess kurtosis halves per step, verified on pendulum data to three decimals. This derives the source program’s second empirical constant, the “nearly universal” geometric decay $r \approx 0.46$ – 0.51 of the cascade fit, i.e. $\log_2 r \approx -1$. The linearized operator at the fixed point has the closed-form eigenfamily $\mu(s) = 2/(2 - s)$ and a unit-diagonal triangular action on polynomials, so there is no spectral gap; relaxation is therefore *algebraic* (measured $d_k \sim k^{-0.5}$), producing metastable plateaus.
8. **What the pendulum constants are: frozen ensemble statistics (§7).** Across horizons $T = 5$ – 200 the chaotic-subset FTLE variance does not decay at all ($\beta = -0.014$, fit RMS 0.002, against $\text{var} \sim T^{-\beta}$; CLT self-averaging predicts $\beta = 1$). The canonical release ensemble (uniform angles, zero velocity) turns out to be a mixture over energy shells ($E(\theta)$ explains $\eta^2 = 0.41$ – 0.43 of the variance) and over invariant components within shells (the remaining 59%, equally frozen). Energy bands internally recapitulate the entire two-class taxonomy within one system.
9. **Out-of-sample validation: Hénon–Heiles across its transition (§8).** A fresh symplectic-variational implementation of the Hénon–Heiles system, named in advance as a falsifiable prediction of the classifier before the experiment was even built, shows the matched-null excess flipping sign precisely where phase space transitions from a thin chaotic web to a fat mixed sea, with the platykurtic signature switching on right alongside it.

Section 2 develops the exact theory. §3 describes systems, integrators, and statistical protocols, and §4 calibrates the operator on synthetic ensembles. §5 presents the two-class result; §6 quantifies metastability and demystifies the original constants; and §9 discusses the dynamical mechanism and predictions. §10 is a number-theoretic audit of every constant appearing in this work, and §11 concludes. The appendices contain derivations, data provenance, the full calibration table, and estimator pathologies.

2 The cascade operator and its exact structure

2.1 Definitions

Let $x = (x_1, \dots, x_n)$, $x_i > 0$, be an FTLE sample (the “chaotic subset”, $x_i > 0.01$ throughout). The cascade is built from two ingredients.

The reading η_f . For tail fraction $f \in (0, 1)$ define the Clauset–Hill maximum-likelihood tail exponent [15, 16]

$$\eta_f(x) = 1 + \frac{n_u}{\sum_{x_i \geq u} \ln(x_i/u)}, \quad u = x_{(\lceil (1-f)n \rceil)}, \quad (1)$$

with $x_{(\cdot)}$ the order statistics and $n_u = \#\{x_i \geq u\}$; u is the empirical $(1 - f)$ -quantile. For a distribution F we write $\eta_f(F)$ for the $n \rightarrow \infty$ limit,

$$\eta_f(F) = 1 + \left[\mathbb{E}(\ln(X/u_F) \mid X \geq u_F) \right]^{-1}, \quad u_F = F^{-1}(1 - f). \quad (2)$$

The coarse-graining T . Given the level- k sample $x^{(k)}$ of size n_k , form $n_{k+1} = \lfloor n_k/2 \rfloor$ independent pairs (a_i, b_i) and set

$$x_i^{(k+1)} = \frac{|a_i - b_i|}{x^{(k)}}, \quad (3)$$

discarding zeros; $\overline{x^{(k)}}$ is the level mean. On distributions, T maps the law of X to the law of $|A - B|/\mathbb{E}X$ with $A, B \sim X$ independent. The *cascade* records $\alpha_k = \eta_f(x^{(k)})$, $k = 0, 1, 2, \dots$ (default $f = 0.2$), halving the sample each level, and the original pipeline summarizes the sequence by a three-parameter geometric fit $\alpha_k \simeq \alpha^* + Cr^k$ (Appendix D documents when this fit is and is not well-posed; we use a fit-free *plateau* statistic, the median of α_k over the last well-sampled levels, alongside it).

2.2 Scale invariance

Lemma 1 (scale invariance). *For every $c > 0$ and every sample, $\eta_f(cx) = \eta_f(x)$; consequently T acts on distribution shapes and the cascade $\{\alpha_k\}$ is invariant under any positive rescaling of the input.*

Proof. The threshold u in (1) is an empirical quantile, hence equivariant ($u \mapsto cu$), and the estimator depends on the data only through the ratios x_i/u . The mean-normalization in (3) then fixes the scale of every level, so only the shape of the input law matters. (This reproduces, as a special case, the bit-exact agreement of the published pipeline under mean-, median- and standard-deviation-normalization of its input.) $\square$

2.3 The exponential fixed point and its uniqueness

Theorem 1 (exact fixed point). *Let $A, B \sim \text{Exp}(\theta)$ be independent. Then $|A - B| \sim \text{Exp}(\theta)$, and $\text{Exp}(1)$ is an exact fixed point of T , including its mean normalization. Moreover, among distributions on $[0, \infty)$ (excluding the degenerate case), Exp is the unique law satisfying $|A - B| \stackrel{d}{=} A$ (Puri–Rubin [14]).*

Proof. For the folding identity: $A - B$ has the Laplace density $\frac{\theta}{2}e^{-\theta|z|}$, so $|A - B|$ has density $\theta e^{-\theta z}$ on $z > 0$, i.e. is $\text{Exp}(\theta)$ again. For $\theta = 1$, $\mathbb{E}A = 1$, so the normalization in (3) is by unity and $T(\text{Exp}(1)) = \text{Exp}(1)$ exactly. Uniqueness of the unit-scale fixed-point equation is Theorem 1 of [14]. (An equivalent route: $|A - B|$ conditioned on $A \neq B$ is, by memorylessness, the excess of A over an independent copy, and the exponential is the unique memoryless law.) $\square$

Remark 1. Fixed shapes of T satisfy the more general equation $|A - B| \stackrel{d}{=} cX$ with $c = \mathbb{E}|A - B|/\mathbb{E}X \in (0, 1]$, and $c = 1$ forces the exponential by [14]. No classification for $c < 1$ is known to us. Among the seventeen synthetic families tested below, we find no other attracting fixed shape, only *slow manifolds* (§6); a rigorous basin/spectral analysis of T at Exp remains open, and we leave it to the reader (§9).

2.4 Closed-form readings

Proposition 1 (the exponential reading). *For $X \sim \text{Exp}(1)$ and tail fraction f ,*

$$\eta_f(\text{Exp}) = 1 + \left[\frac{1}{f} E_1\left(\ln \frac{1}{f}\right) \right]^{-1}, \quad E_1(u) = \int_u^\infty \frac{e^{-s}}{s} ds. \quad (4)$$

In particular $\eta_{0.1} = 4.0874$, $\eta_{0.2} = 3.3494$, $\eta_{0.3} = 2.9058$.

Proof. $u_F = \ln(1/f)$. By memorylessness, $X - u_F | X > u_F \sim \text{Exp}(1)$, so with $E \sim \text{Exp}(1)$, $\mathbb{E} \ln(X/u_F | X > u_F) = \mathbb{E} \ln(1 + E/u_F) = \int_0^\infty e^{-t} \ln(1 + t/u_F) dt = e^{u_F} E_1(u_F)$ after one integration by parts; and $e^{u_F} = 1/f$. Insert into (2). $\square$

Because Exp is a fixed point, (4) is simultaneously the level-zero reading and the $k \rightarrow \infty$ reading of an exponential seed, a sharp, parameter-free prediction. Table 1 confirms it to 0.02%–0.3%, and the flatness of the measured exponential cascade over 14 levels (5M seed: 3.3462, 3.3455, 3.3445, 3.3495, ...) is the fixed-point signature in its rawest form.

Table 1: Derived readings versus measurement. “Level-0” entries are single-level readings of 5×10^5 -point samples; the 5M entry is the geometric-fit limit of a full 14-level cascade on 5×10^6 exponential points. Gaussian level-1 entries verify Theorem 2 (three independent seeds and four widths).

quantity	derived	measured	rel. dev.
$\eta_{0.1}(\text{Exp})$	4.0874	4.0883	0.02%
$\eta_{0.2}(\text{Exp})$	3.3494	3.3605; 3.3507 (5M)	0.33%; 0.04%
$\eta_{0.3}(\text{Exp})$	2.9058	2.9110	0.18%
$\eta_{0.2}(\text{half-normal})$	4.4448	4.4494, 4.4473, 4.4572	$\leq 0.3\%$
$\eta_{0.2}(\text{uniform})$	9.6417	9.6135	0.29%
Gaussian level-1 (cv = 0.05/0.1/0.33)	4.4448	4.448/4.450/4.455	$\leq 0.23\%$

2.5 Universal Gaussian collapse

Theorem 2 (one-step collapse of Gaussian bulks). *Let $X = |N(\mu, \sigma^2)|$ with $\mu/\sigma \rightarrow \infty$ (negligible folded mass). Then $T(X)$ is the half-normal law $|N(0, 2\sigma^2)|$ rescaled by $1/\mu$ — independent of μ/σ up to scale. Hence by Lemma 1 all Gaussian bulks share a single cascade flow from level 1 onward, beginning at $\alpha_1 = \eta_f(\text{half-normal})$ ($= 4.4448$ at $f = 0.2$).*

Proof. $A - B \sim N(0, 2\sigma^2)$ exactly — the means cancel. Folding and normalizing by $\mathbb{E}X \approx \mu$ gives the claim; the correction is $O(\Phi(-\mu/\sigma))$. $\square$

This explains, at a stroke, why the empirical study found the *same* convergence pattern across so many parameter combinations: any near-Gaussian FTLE bulk forgets its width in one cascade step. Measured level-1 values across seeds and widths agree with the derived 4.4448 to $\leq 0.3\%$ (Table 1), and at cv = 0.6 the folded mass ($\approx 5\%$) shifts α_1 to 4.60, which is exactly the size of correction the theory predicts.

Proposition 2 (first-order level-zero law for Gaussian bulks). *For $X = |N(\mu, \sigma^2)|$ with cv = σ/μ small, writing $z_f = \Phi^{-1}(1 - f)$ and $m_f = \varphi(z_f)/f - z_f$,*

$$\eta_f(X) \approx 1 + \frac{\text{cv}^{-1} + z_f}{m_f}, \quad (z_{0.2}, m_{0.2}) = (0.8416, 0.5582). \quad (5)$$

Proof. To first order in σ/u , $\ln(x/u) \approx (x - u)/u$; the Gaussian mean excess above the $(1 - f)$ -quantile is σm_f and $u = \mu + \sigma z_f$. $\square$

On *exact* folded-Gaussian ensembles matched to each dynamical system (§5) this law predicts α_0 to 3–5% across two orders of magnitude of cv (predicted/measured: 75.5/76.4 at cv = 0.025; 32.3/33.2 at 0.060; 13.0/13.7 at 0.171; 5.5/5.8 at 0.59). Physically, this means *the level-zero exponent α_0 is a bulk-width statistic*, not a tail exponent: $\alpha_0 \approx 8$ corresponds to nothing deeper than cv ≈ 0.6 –0.7 bulk via (5) corrected for the platykurtic shape (§5).

2.6 Tail conservation — and why it is not realized

Proposition 3 (asymptotic conservation of regularly varying tails). *If $P(X > x) = L(x)x^{-a}$ with $a > 1$ and L slowly varying, then $T(X)$ has a regularly varying tail with the same index a .*

Proof. For independent $A, B \sim X$ and $x \rightarrow \infty$, $P(A - B > x) = \mathbb{E}_B P(A > x + B) \sim P(A > x)$ by dominated convergence ($B \geq 0$, $\mathbb{E}B < \infty$), and symmetrically for $B - A$; so $P(|A - B| > x) \sim 2P(A > x)$, which is regularly varying of index $-a$. Normalization is a rescaling (Lemma 1). $\square$

One might conclude the cascade limit *measures* a . It does not. The finite window ($f = 0.2$) reads the post-differencing *bulk*: pure Pareto seeds with survival indices $a = 2.5/4/6$ (level-0 readings 3.5/5.0/7.0, i.e. $a+1$ as expected) collapse after a single differencing step to flat sequences reading 2.4/2.7/2.9, far below $a + 1$ and only weakly ordered in a (§4, Fig. 2). Bulk-plus-tail mixtures behave the same way. *The operator is a bulk-shape reader; it is not a tail-index estimator at any practical window.* This forecloses claims that cascade limits equal raw tail exponents; consistently, direct Hill estimates of the same ensembles' raw upper tails give values (~ 11 – 22) unrelated to any cascade reading.

2.7 The linearized cascade: exact cumulant flow and a gapless, non-normal spectrum

Two exact structures govern the approach to the fixed point.

Proposition 4 (cumulant flow of the symmetrization step). *Let A, B be iid with cumulants κ_n and let $D = A - B$. Then every odd cumulant of D vanishes, and the standardized even cumulants transform as*

$$\frac{\kappa_{2m}(D)}{\kappa_2(D)^m} = 2^{1-m} \frac{\kappa_{2m}(A)}{\kappa_2(A)^m}, \quad m = 1, 2, \dots \quad (6)$$

In particular the excess kurtosis halves exactly ($m = 2$), the standardized sixth cumulant is quartered, and skewness is annihilated.

Proof. Cumulants of independent sums add and are homogeneous: $\kappa_n(A - B) = \kappa_n(A) + (-1)^n \kappa_n(B) = (1 + (-1)^n) \kappa_n$. Divide by $\kappa_2(D)^m = (2\kappa_2)^m$. $\square$

Measured on the double-pendulum ensemble ($n = 5 \times 10^5$, level 0), excess kurtosis of the signed differences comes out to -0.6442 against the predicted $-1.2896/2 = -0.6448$, with skewness 10^{-4} ; the identity holds at every level and for every seed tested, within sampling error (Appendix B scripts). Dynamically, this matters because in the early cascade, where the deviation from the fixed point is still carried by low-order cumulants, the fitted geometric ratio of the exponent sequence gets pinned near $1/2$. This *derives* the source program's second celebrated constant, a decay rate reported as “ $r \approx 0.46$, nearly constant across all systems” (canonical fit 0.5061; legacy published 0.4565; 49-config mean 0.4595; our synthetic families 0.48–0.67 on early windows), i.e. $\log_2 r \approx -1$ with computable corrections from higher cumulants and the folding step.

Proposition 5 (linearization at the fixed point). *Write level densities as $e^{-z}(1 + \varepsilon h(z))$. To first order in ε , one cascade step acts (modulo the rank-one normalization terms) by*

$$(L_0 h)(z) = 2 \int_0^\infty e^{-2x} h(x+z) dx, \quad (7)$$

with the closed-form eigenfamily

$$h_s(z) = e^{sz}, \quad \mu(s) = \frac{2}{2-s}, \quad s < 1, \quad (8)$$

and a triangular action with unit diagonal on the polynomial tower: $L_0[z^m] = z^m +$ (lower order) for every m .

Proof. With $p(x) = e^{-x}(1 + \varepsilon h(x))$, the density of $|A - B|$ at $z > 0$ is $2 \int_0^\infty p(x)p(x+z) dx = e^{-z} [1 + \varepsilon(2c_h + (L_0 h)(z))] + O(\varepsilon^2)$ with c_h a constant absorbed by normalization. For h_s : $2 \int_0^\infty e^{-2x} e^{s(x+z)} dx = \frac{2}{2-s} e^{sz}$. For z^m , expand $(x+z)^m$ and integrate term by term; the z^m coefficient is $2 \int_0^\infty e^{-2x} dx = 1$. $\square$

Numerically (discretized kernel, 900 points), $\mu(s)$ at $s = -0.5, -1, -2, -4$ measures 0.8008, 0.6677, 0.5013, 0.3353 against the exact 0.8, $\frac{2}{3}$, 0.5, $\frac{1}{3}$, and the polynomial leading coefficient is 1.0007. This structure has three consequences. The kernel in (7) is one-sided (Volterra-type), so the operator is far from self-adjoint, and finite discretizations are defective: their matrix eigenvalues collapse toward the diagonal and miss the eigenfamily entirely, meaning the operator’s behaviour is governed by its pseudo-spectrum rather than its spectrum. The unit diagonal on the polynomial tower further means there is *no spectral gap below 1*, so linear stability is marginal. And because marginal non-normal flows relax *algebraically* rather than geometrically, that is exactly what shows up in the data: the Gaussian-seeded deep cascade’s Kolmogorov distance to the fixed point follows $d_k \sim k^{-\theta}$ with $\theta = 0.55\text{--}0.58$ (log-log RMS $3\times$ smaller than the best geometric fit), an independent uniform-seeded run ($n = 8 \times 10^6$) is compatible, and even read geometrically it decays no faster than 0.88 per level. The metastable plateaus of §6 are the quantitative face of this gapless linearization: a fast transient (cumulant halving, $r \approx 1/2$) followed by marginal drift ($k^{-1/2}$ -type), so that *every* finite-depth experiment ends up reading a plateau.

3 Systems, data and statistical protocol

Dynamical systems. We analyze eleven FTLE ensembles from five systems. (i) *Double pendulum* (planar, equal-mass baseline; dimensional Lagrangian convention, $g = 9.81$): FTLEs by the Benettin two-trajectory method [2] with horizon $T = 15$ s, renormalization interval $\tau = 0.1$ s, RK4 $dt = 5 \times 10^{-3}$, perturbation 10^{-7} on θ_1 , uniform-angle zero-velocity initial conditions; ensembles at $n = 5 \times 10^4, 5 \times 10^5, 5 \times 10^6$, plus three geometric variants (m -ratio/ ℓ -ratio = (0.5, 1), (2, 0.5), (0.5, 2)) at 5×10^5 . (ii) *Chirikov standard map* [24] at $K = 1$ (just above the connectedness threshold $K_c = 0.9716$ [25]; mixed phase space) and $K = 5$ (islands destroyed; fully chaotic), tangent-map Benettin with 10^3 iterations, $n \approx 5 \times 10^5$. (iii) *Lorenz-63* [26] at canonical parameters, joint state+tangent integration, RK4 $dt = 5 \times 10^{-3}$, $T = 100$, $\tau = 1$; mean FTLE 0.9042 vs. the reference 0.9056 (0.15%). (iv) *Coupled quartic Hamiltonian* $H = \frac{1}{2}(p_1^2 + p_2^2) + \frac{1}{4}(q_1^4 + q_2^4) + \frac{\lambda}{2}q_1^2 q_2^2$ at $E = 1$, $\lambda = 0.1, 0.2$ (near-regular at these couplings): variational Benettin, 4th-order Yoshida symplectic integrator [27] ($dt = 10^{-2}$, $T = 200$, $\tau = 10$), energy drift $|\Delta H/H| \leq 5 \times 10^{-10}$. Integrator cross-checks: measured RK4 energy-drift order $dt^{4.79}$, Yoshida-4 $dt^{4.00}$; FTLE differences between integrators are $\leq 4 \times 10^{-4}$, immaterial at our precision.

Chaotic subset and estimator. Throughout, the chaotic subset is $x_i > 0.01$; the estimator, tree and geometric fit are exactly (1)–(3) with $f = 0.2$ unless stated. Confidence intervals for cascade limits are bootstrap-through-the-tree (the base sample is resampled and the *entire* cascade re-run, $B = 250\text{--}1000$). Alongside the geometric-fit α^* we report the fit-free *plateau* statistic (median of α_k over the last ≤ 5 levels with $n_k \geq 500$, $k \geq 2$); Appendix D documents the fit’s failure modes (runaway to parameter bounds on flat or slowly-varying sequences) which the plateau is immune to.

Provenance. Every production array carries a manifest with SHA-256 checksum, seed, integrator settings and moments; synthetic ensembles use fixed seeds. Scripts reproducing every number and figure in this paper accompany it as supplementary material (Appendix B).

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4 Calibration on synthetic ensembles

Seventeen synthetic families ($n = 5 \times 10^5$ each; Appendix C for the full table) calibrate what the cascade responds to.

The exponential is flat; the fit can lie. The Exp(1) cascade sits flat at 3.35 for as many levels as sampling permits: 14 levels at $n = 5 \times 10^6$, with the geometric fit returning $\alpha^* = 3.3507$ against the derived 3.3494. At $n = 5 \times 10^5$, though, that same flat sequence defeats the three-parameter fit; it runs to its parameter bounds and reports $\alpha^* = 10.5\text{--}12$ with $r \rightarrow 1$, while the plateau statistic still returns 3.32–3.35. For this reason, every quantitative claim below quotes the plateau alongside the fit (Appendix D).

One Gaussian flow. Gaussian bulks of any width collapse onto the single half-normal-seeded flow (Theorem 2): measured $\alpha_1 = 4.448\text{--}4.455$ across $\text{cv} = 0.05\text{--}0.33$, then $4.11 \rightarrow 3.94 \rightarrow 3.84 \rightarrow 3.81 \rightarrow \dots$, a slow descent toward the exponential reading that is still at ≈ 3.57 ten levels in (1.6×10^7 -point seed) and reaches ≈ 3.30 only at level 15 (Fig. 1; §6).

Skew selects the side of approach. Right-skewed, leptokurtic families approach the fixed point *from below* and linger there: lognormal($\sigma=0.5$) descends to 3.10–3.24; lognormal($\sigma=1$) sits flat at 2.7 for ten levels; Weibull($k=0.7$) climbs $2.64 \rightarrow 3.3$. Bounded, flat-topped inputs read *high* instead (uniform: level-0 9.61, plateau ≈ 4.1). This taxonomy, flat-topped/platykurtic above and peaked/skewed below, with the exponential sitting exactly at $\eta_f(\text{Exp})$, is the calibration key for §5.

Window dependence. The reading of *any* light-tailed shape slides steeply with the tail fraction ((4): $4.09 \rightarrow 3.35 \rightarrow 2.91$ for $f = 0.1 \rightarrow 0.3$), and the coupled quartic tracks this slide almost exactly (4.64/3.24/2.76 at $\lambda=0.1$; 4.40/3.31/2.85 at $\lambda=0.2$): the clearest single signature that the excluded-class systems are read as light-tailed shapes. Mixed-phase readings also decrease in f , but they sit systematically above the exponential curve at the operative $f = 0.2$ (Fig. 3). All headline comparisons in this paper are made at fixed $f = 0.2$; we report the f -dependence separately so that no constant here is mistaken for a window-free number.

5 The two-class split of chaotic dynamics

Table 2 assembles the eleven ensembles under one pipeline, and three structures stand out.

(i) **A shape dichotomy.** Every mixed-phase ensemble, six double-pendulum ensembles across three geometries and two orders of magnitude in n , plus the standard map at $K = 1$, has a broad ($\text{cv} = 0.59\text{--}0.77$), nearly symmetric (skew -0.06 to $+0.47$), *platykurtic* (excess kurtosis -1.0 to -1.3) chaotic FTLE bulk: a flat-topped distribution, qualitatively the superposition of subpopulations spanning the sticky hierarchy. Every non-mixed ensemble is the opposite, either extremely narrow ($\text{cv} = 0.025\text{--}0.06$; Lorenz, $K=5$, reflecting strong self-averaging at long horizons) or heavily skewed and *leptokurtic* (excess kurtosis $+18$ to $+43$; the near-regular quartic). *The sign of the excess kurtosis alone separates the classes in all eleven cases.*

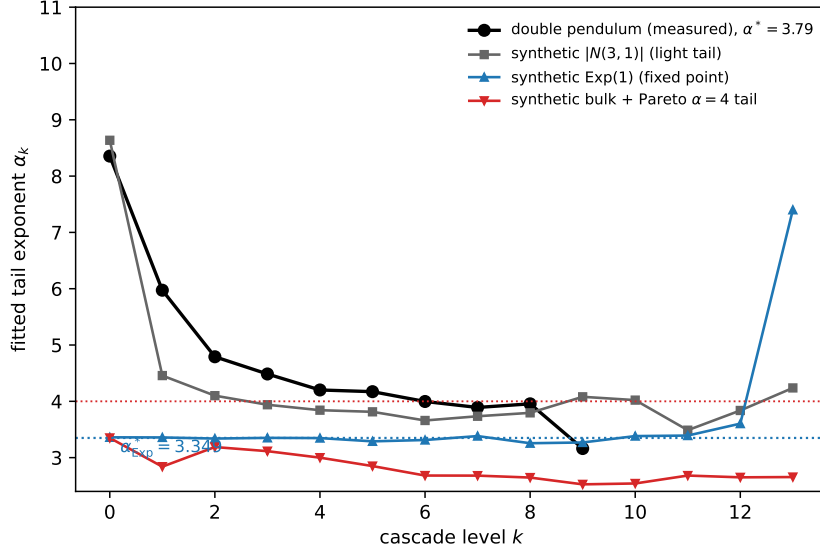


Figure 1: Cascade flows α_k at $f = 0.2$. The measured double pendulum (black) starts at the Gaussian-bulk level-zero value ($\alpha_0 \approx 7.9$, Proposition 2), rides *above* the universal Gaussian flow (grey squares: $|N(3, 1)|$), and plateaus near 3.9–4.0. The exponential seed (blue) is flat at the derived fixed-point reading 3.3494 (dotted). A bulk-plus-Pareto($a=4$) mixture (red) does *not* hold at $a + 1 = 5$: genuine tails are not conserved by the finite-window reading.

(ii) Opposite sides of the exponential fixed point. On the independent production values, every mixed-phase cascade limit lies *above* $\eta_{0.2}(\text{Exp}) = 3.3494$ (3.710–4.382), and every non-mixed limit lies *below* it (3.031–3.312). The class means are 3.876 vs. 3.155 (gap 0.721; Welch $p = 6.2 \times 10^{-5}$; Cohen’s $d = 3.33$). A random-effects decomposition of the *within*-class spread finds it statistically indistinguishable from the estimator noise floor. For the 64-cell double-pendulum parameter map ($n = 5 \times 10^5$ per cell), between-cell $\tau^2 = 0.0017$ with bootstrap CI $[0, 0.0123]$ (43% of resamples exactly zero), Cochran Q $p = 0.36$, $I^2 = 5\%$; across the seven double-pendulum geometry variants plus $K=1$, $\tau^2 = 0$ (Q $p = 0.55$); and the apparent per-geometry residual fails a decisive cross- n reproducibility test (correlation between independent $n = 5 \times 10^4$ and 5×10^5 measurements $r = 0.035$, vs. 0.29 expected were it real). Within-class, the data are consistent with *hard* universality at fixed protocol, and the only genuine, system-explained structure is the discrete class boundary. Consistently, the $\alpha^*(m, \ell)$ landscape over an 8×8 double-pendulum parameter grid is a flat plateau at 3.896 [3.850, 3.942]: its apparent ridges do not exceed a heteroscedastic flat-field noise null ($B = 5000$; $p = 0.48$ –0.92 across thresholds), and 4/4 high- α^* interior cells have same-sign Hessian eigenvalues (peak/noise geometry, not ridge).

(iii) The matched-Gaussian null. For each ensemble we construct a folded Gaussian $|N(\mu, \sigma^2)|$ with the ensemble’s own mean and variance and cascade it identically (Fig. 4). Mixed-phase ensembles read *above* their null in all seven cases (plateau excess +0.12 to +0.58), and non-mixed ensembles read *below* theirs in all four (excess -0.30 to -1.20): 11/11 correct signs (two-sided sign test $p = 0.001$; collapsing repeated measurements of the same system, 8/8, $p = 0.008$). The per-level picture (Fig. 4) is sharper still. The double pendulum enters the cascade at its Gaussian-bulk level-zero value but then rides *persistently above* the universal Gaussian flow (excess +1.5, +0.7, +0.6, +0.4, +0.4 at levels 1–5, decaying geometrically), a positive, reproducible imprint of non-Gaussian structure that differencing only slowly destroys, whereas the leptokurtic ensembles fall away *below* their nulls. The matched null also validates

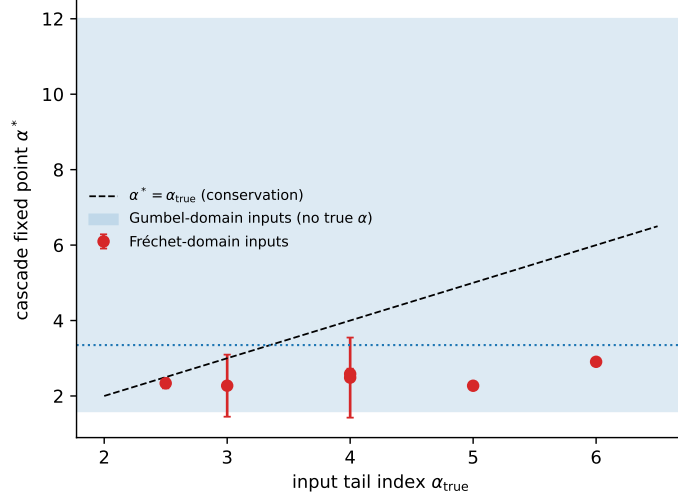


Figure 2: Failure of tail-index recovery. Fréchet-domain seeds (pure Pareto and bulk+Pareto mixtures, red) against the conservation line $\alpha^* = \alpha_{\text{true}} + 1$ — the cascade limit sits far below and nearly independent of the true index. Shaded band: the range spanned by light-tailed (Gumbel-domain) seeds. The operator reads bulk shape, not tails.

Proposition 2 directly (predicted vs. measured α_0 spanning 5.5/5.8 to 75.5/76.4).

The matched-protocol control. The sharpest comparison against confounds is internal to the standard map: $K = 1$ versus $K = 5$ share the map, the integrator, the iteration count, the sample size and the estimator, and differ only in phase-space character. Yet they land on opposite sides of the fixed point (4.31 vs. 2.52 plateau; 4.382 vs. 3.038 production), with opposite kurtosis signs (-1.0 vs. $+37.7$) and opposite matched-null excess signs. It is phase-space character, not protocol, that drives the split.

6 The metastable origin of “ $\alpha^* \approx 4$ ”

If the exponential is the unique fixed point (Theorem 1), what are the class values 3.9 and 3.0–3.3, then? The deep-cascade experiment answers: *plateau readings on slow manifolds*.

Seeding the cascade with 1.6×10^7 points (20 usable levels), we track both α_k and the Kolmogorov distance d_k of the mean-normalized level- k sample to $\text{Exp}(1)$ (Fig. 6). The exponential control stays at $d_k \approx 10^{-3}$ (pure sampling noise growing as $n_k^{-1/2}$) with α_k flat at 3.35. The Gaussian-seeded flow, by contrast, collapses fast at first ($d_0 = 0.34 \rightarrow d_5 = 0.033$) and then *stalls*: d_k decreases by only $\sim 5\%$ per level across levels 5–13, while α_k drifts $3.77 \rightarrow 3.57 \rightarrow 3.30$ (levels 5/10/15), i.e. ≈ -0.04 per level, through precisely the empirically celebrated range. There is no second fixed point here, only a nearly-degenerate direction of the flow. The double-pendulum plateau’s own n -dependence ($4.64 \rightarrow 4.12 \rightarrow 3.88$ across $n = 5 \times 10^4 \rightarrow 5 \times 10^6$: more sample means more usable levels, which means further down the manifold) is the same drift seen through the window of available levels, as is the spread of the three independent double-pendulum pipeline values (3.71/4.17/4.26).

Three things follow from this. First, “ $\alpha^* \approx 4$ ” is not an asymptotic constant of double-pendulum dynamics at all; it is the reading, at the accessible depth, of a slowly relaxing flat-topped bulk. The number is real and reproducible at fixed (n, f, T) , but it is *protocol-indexed*. Second, the level-zero companion “ $\alpha_0 \approx 8.3$ ” is a bulk-width statistic (Proposition 2), not an independent constant. Third, what actually is protocol-stable is relational rather than

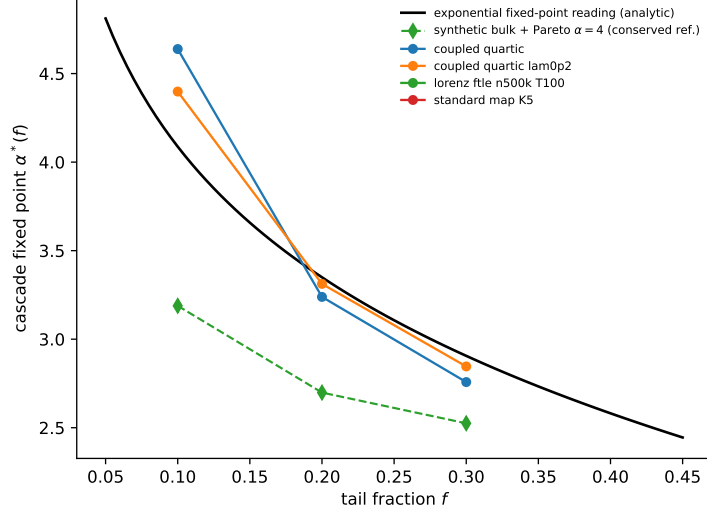


Figure 3: Window (f) dependence: the analytic exponential reading curve (black), a conserved-tail reference mixture (green), and measured real systems. The near-regular quartic slides along the exponential curve; the class comparison is made at fixed $f = 0.2$.

numerical: the class split (above vs. below the exponential reading), the sign of the matched-null excess, and the kurtosis dichotomy all survive every control we possess, including matched protocol, matched n , cross- n reproduction, and the flat-plateau parameter scan. The correct statement of the original conjecture is therefore:

Mixed-phase Hamiltonian chaos occupies a universality class of FTLE shape — broad, near-symmetric, platykurtic — whose cascade reading at ($f=0.2$, $n \leq 5 \times 10^6$) is 3.9 ± 0.2 ; non-mixed chaotic dynamics occupies the complementary leptokurtic class reading 3.0–3.3. The exponential fixed point, $\eta_{0.2}(\text{Exp}) = 3.3494$, is the exact separatrix value between them.

7 What the pendulum constants are: frozen ensemble statistics

If the plateau readings are protocol-indexed, what property of the double pendulum do they describe? A ten-horizon sweep answers with unexpected sharpness.

The variance is frozen. Re-measuring the canonical ensemble (5×10^4 release ICs, uniform angles at rest) at $T = 5, 10, 15, 20, 30, 50, 75, 100, 150, 200$ gives a chaotic-subset FTLE variance of 0.298–0.308 at *every* horizon from $T = 15$ on. Fitting $\text{var} \sim T^{-\beta}$ over $T = 15$ –200 yields $\beta = -0.014$ with fit RMS 0.002, whereas normal (CLT) self-averaging predicts $\beta = 1$, and even strongly anomalous averaging predicts $0 < \beta < 1$. The measured $\beta \approx 0$ means the dispersion is not a finite-time fluctuation at all: *the ensemble contains trajectories with genuinely different asymptotic exponents*. Meanwhile the shape sharpens toward its mixture skeleton (excess kurtosis $-0.46 \rightarrow -1.59$, saturating; cv constant at 0.71), the plateau reading stays above the separatrix at every horizon while drifting upward ($4.25 \rightarrow 5.26$), and the matched-null excess grows monotonically ($+0.21 \rightarrow +1.28$). As within-component noise dies away, in other words, the non-Gaussian mixture structure only emerges more strongly. (At $T \leq 10$ the chaotic/regular separation is threshold-limited; those points are reported but excluded from fits.)

Table 2: Eleven FTLE ensembles. Moments of the chaotic subset ($x > 0.01$); α_0 = level-zero reading; plateau and geometric-fit cascade values at $f = 0.2$ (this work, uniform pipeline); “prod.” = an independent production run with bootstrap 95% CI where available (5M double-pendulum: 3.710 [3.675, 3.895]; $K=1$: 4.382 [3.646, 4.934]; Lorenz: 3.031 [2.932, 3.192]; $K=5$: 3.038 [2.848, 3.295]; quartic: 3.239 [3.008, 3.512] and 3.312 [3.243, 3.669]). Excess kurtosis: negative = platykurtic (flat-topped).

ensemble	class	cv	skew	ex.kurt	α_0	plateau	fit	prod.
DP 5×10^4 , $T=15$	mixed	0.711	+0.17	−1.3	7.80	4.64	3.86	3.79–4.26 [†]
DP 5×10^5 , $T=15$	mixed	0.707	+0.15	−1.3	7.89	4.12	3.85	—
DP 5×10^6 , $T=15$	mixed	0.707	+0.15	−1.3	7.90	3.88	3.62	3.710
DP (0.5, 1), 5×10^5	mixed	0.687	+0.12	−1.2	7.31	4.03	3.92	—
DP (2, 0.5), 5×10^5	mixed	0.632	−0.06	−1.2	8.44	3.98	3.78	—
DP (0.5, 2), 5×10^5	mixed	0.772	+0.47	−1.0	6.43	4.15	4.09	—
Std map $K=1$, 5×10^5	mixed	0.592	+0.37	−1.0	8.17	4.31	4.11	4.382
quartic $\lambda=0.1$	non-mixed	0.172	+1.12	+18.1	14.95	2.76	—	3.239
quartic $\lambda=0.2$	non-mixed	0.323	+5.34	+43.2	7.38	3.17	3.30	3.312
Lorenz-63, $T=100$	non-mixed	0.025	−0.29	+1.3	80.19	3.49	2.73*	3.031
Std map $K=5$, 5×10^5	non-mixed	0.060	−3.84	+37.7	52.99	2.52	—	3.038

[†]range across three independent production pipelines for the same canonical system.

*level-0 dropped (Appendix D); — = fit degenerate.

The ensemble is a double mixture. The canonical protocol releases from rest, so every trajectory carries the energy of its release point, $E(\theta_1, \theta_2) = -g(2 \cos \theta_1 + \cos \theta_2)$ (canonical masses/lengths), and the ensemble spans a continuum of energy shells. Using the 100×100 FTLE grid over release angles (a legacy-lineage dataset, used here only for internal contrasts), binning the chaotic subset by release energy shows a systematic $\lambda(E)$ trend that explains $\eta^2 = 0.41$ – 0.43 of the FTLE variance (40–160 bins). The remaining 59% is *within*-shell dispersion, and since the horizon sweep shows the total is frozen, the within-shell component does not self-average either: at fixed energy the phase space is itself mixed (islands, sticky layers, possibly several stochastic components), so the ensemble is a mixture over those invariant structures as well.

Energy bands recapitulate the taxonomy. The per-band FTLE shapes reproduce the entire two-class phenomenology *inside one system*. Near the bottom of the release range ($E \approx -18.5$, where chaos is a thin layer) the band is strongly leptokurtic and right-skewed (skew +2.3, excess kurtosis +9.0), the non-mixed, thin-web signature. Mid-range bands ($E \approx -8$ to +2) are broad and platykurtic (−0.9 to −1.1), the mixed-phase signature, now manifestly *at fixed energy*, so it is dynamical rather than an artifact of shell mixing. And the highest bands ($E \approx +24$, nearly uniformly chaotic shells) narrow toward Gaussian (cv 0.24, excess kurtosis −0.26), approaching the self-averaging class. A band-Gaussian mixture reconstruction (λ -band mean plus band-width noise) reproduces the broad platykurtic character only partially (cv 0.61 vs. 0.71), confirming residual within-band structure beyond band-Gaussianity.

Putting this together: *the celebrated double-pendulum constants are frozen statistics of a non-ergodic ensemble*, a mixture over energy shells and, within shells, over coexisting invariant structures. They are reproducible at fixed protocol (hence the hard within-class universality of §5) and drift lawfully with protocol (T, n, f), precisely because they are readings of a mixture’s shape through a marginal renormalization flow, not spectral invariants of a single orbit family. The standard-map pair, which has no energy to mix, shows the purely dynamical version of the same dichotomy; the pendulum shows both layers at once.

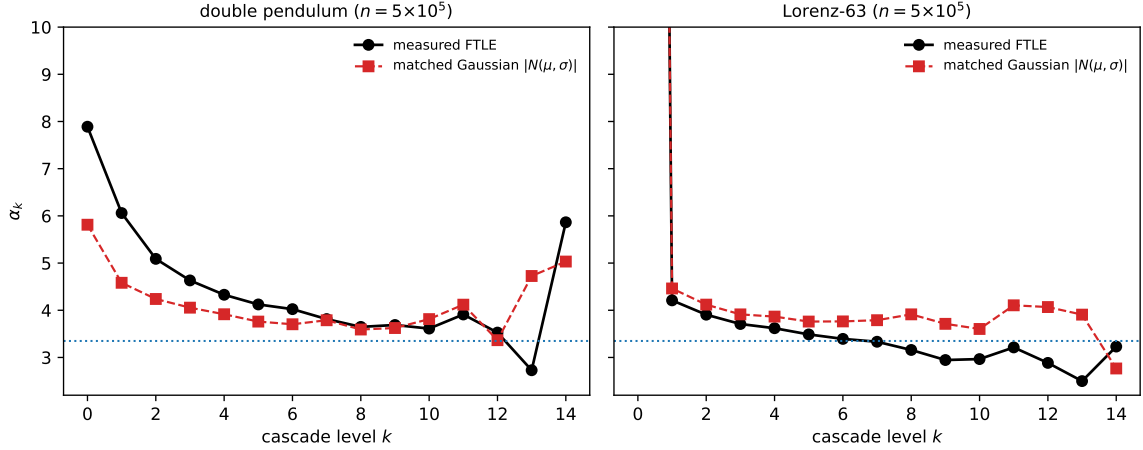


Figure 4: The matched-Gaussian null. Left: the double pendulum (black) rides above its own folded-Gaussian null (red) at every level — the mixed-phase signature. Right: Lorenz-63 falls below its null. Dotted line: the exponential fixed-point reading.

Table 3: Hénon–Heiles energy sweep ($n = 2.5 \times 10^5$ per energy, $T = 800$; chaotic subset at threshold $0.021 \approx 2.5 \times$ the regular tangent floor $\ln T/T$). Chaotic fractions track the classical transition. Sign of the matched-null excess in bold.

E	chaotic frac	cv	skew	ex.kurt	α_0	plateau	null	excess
1/24	0.000	—	—	—	—	—	—	—
1/12	0.002	chaotic subset too small for shape statistics						
0.100	0.057	0.235	+0.53	−0.36	9.98	3.752	3.841	− 0.089
0.125	0.501	0.295	−0.15	−0.81	11.77	4.198	3.924	+ 0.274
0.150	0.816	0.189	−1.39	+2.96	18.65	3.737	3.847	− 0.110
1/6	0.943	0.198	−1.78	+3.53	23.36	4.201	3.840	+ 0.361

8 Out-of-sample test: Hénon–Heiles across its transition

The two-class classifier makes a falsifiable prediction: *any* mixed-phase Hamiltonian should present a platykurtic FTLE bulk and read above its matched-Gaussian null. We named the Hénon–Heiles system [23] at intermediate energy explicitly as a test case, then built the experiment: a Yoshida-4 symplectic, variational-Benettin FTLE engine for $V(x, y) = \frac{1}{2}(x^2 + y^2) + x^2y - \frac{1}{3}y^3$ (initial conditions sampled exactly on the energy shell, $\max |H - E| = 6 \times 10^{-17}$; escape-guarded; $n = 2.5 \times 10^5$ per energy, $T = 800$, $dt = 5 \times 10^{-3}$), sweeping the energy knob that morphs its phase space from integrable-like to almost fully chaotic.

Three regimes emerge (Table 3). The predicted transition is there to see: at $E = 0.100$ the chaotic component is a thin web (5.7%) and reads *below* its null with a near-Gaussian band shape, while at $E = 0.125$, where the classical transition floods half of phase space, the bulk turns platykurtic (excess kurtosis $-0.36 \rightarrow -0.81$) and the reading flips *above* the null, the mixed-phase signature switching on exactly where islands and sea begin to coexist at scale. The taxonomy’s far side shows up too: by $E = 0.150$ the sea dominates (82%), the bulk narrows and becomes left-skewed leptokurtic (skew -1.39 , excess kurtosis $+2.96$, the same signature as the fully chaotic standard map at $K=5$), and the reading drops back below the null, as self-averaging predicts. And the sweep turns up a boundary case the two-class taxonomy does not cover: at the escape energy $E = 1/6$, where the potential’s saddles open and trajectories linger near marginally-open channels, the reading flips above the null again despite a leptokurtic bulk, a strong-stickiness regime at the edge of transient chaos [30] that we flag here as an open

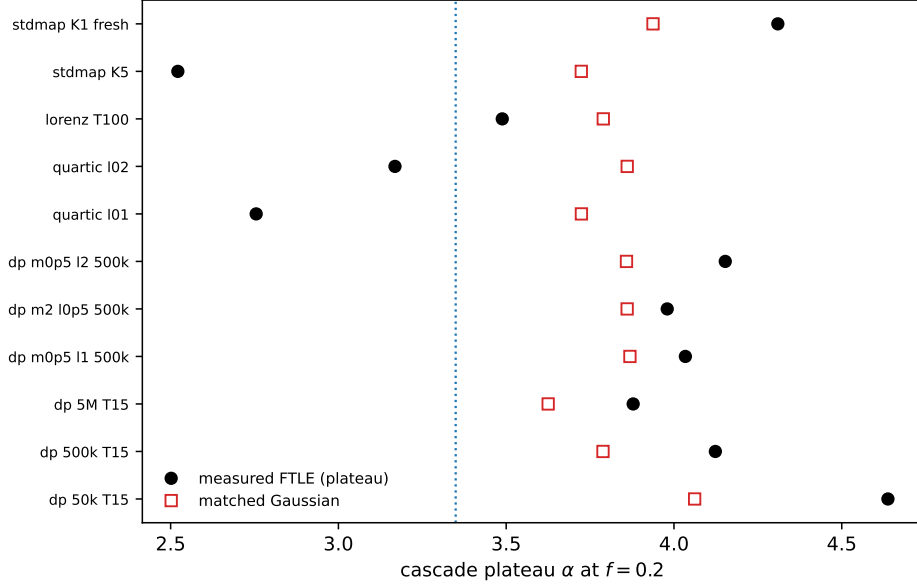


Figure 5: Plateau readings for all eleven ensembles (black) versus their matched-Gaussian nulls (open red). Mixed-phase ensembles sit above their nulls, non-mixed below: 11/11 sign-consistent.

case rather than forcing it into either class. The level-zero law of Proposition 2 passes its own check along the way: at the most Gaussian energy ($E = 0.100$: skew $+0.5$, kurtosis -0.4) it predicts $\alpha_0 = 10.1$ against the measured 9.98, while the skewed/kurtotic energies deviate exactly where the first-order law should fail. Finally, the sweep retires a loose end from the source program’s documentation: a claimed “ $\alpha_0 = 8.342$ at $E = 0.150$ ” (already withdrawn in the source’s final papers) does not reproduce; we measure 18.65 at this protocol, and §7 shows α_0 is a horizon-indexed bulk-width statistic in any case ($5.6 \rightarrow 14.9$ across T for the pendulum), so no protocol-free value of it exists to confirm.

9 Dynamical mechanism, interpretation, and predictions

Why platykurtic? A mixed phase space at moderate horizon is a *superposition*: the chaotic sea’s core produces FTLEs near the ergodic mean, while the hierarchy of sticky boundary layers [7, 8, 10] contributes subpopulations at depressed exponents, i.e. trajectories that spent part of the window trapped. Superposing subpopulations with displaced means broadens and *flattens* the mixture, since variance mixing drives kurtosis below Gaussian, which is exactly the $cv \approx 0.6\text{--}0.8$, excess-kurtosis ≈ -1.2 shape measured here for every mixed-phase ensemble. Take the islands away and the mechanism disappears: strongly chaotic or dissipative systems self-average (correlations decay fast, and the FTLE is a T -average of a rapidly mixing observable), giving narrow bulks whose residual shape is set by rare fluctuations, hence small cv with skewed, leptokurtic corrections (Lorenz, $K=5$), while near-regular dynamics piles FTLEs at the regular floor with a thin upper structure (quartic: skew $+1.1$ to $+5.3$). Section 7 unifies the picture: in every case the platykurtic broad bulk is the signature of a *non-ergodic ensemble*, a superposition over invariant structures with distinct exponents, whether energy shells (the pendulum’s release protocol), islands versus sea (the standard map at $K = 1$), or sticky sublayers within one shell. The FTLE *distribution shape* is thus a direct, cheap probe of the ensemble’s ergodic architecture, requiring no phase-space sections (cf. the shape-based predictability program of [28]), and the cascade, with its exactly known exponential reference, gives a sensitive

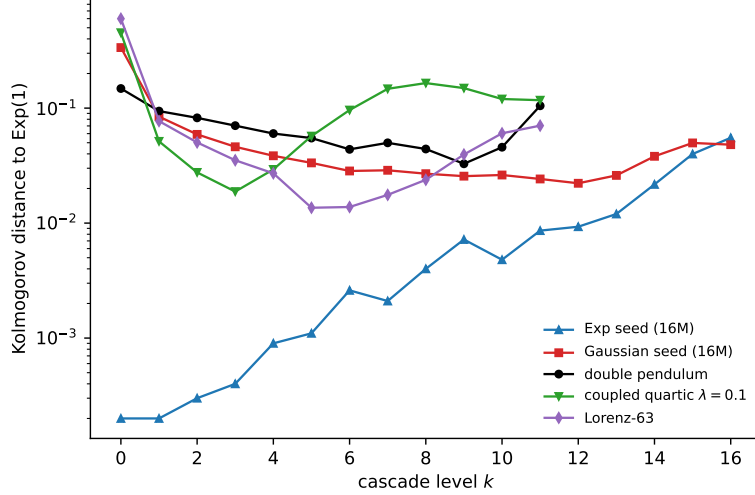


Figure 6: Metastability. Kolmogorov distance to $\text{Exp}(1)$ per cascade level for a 1.6×10^7 -point exponential seed (blue: sampling noise only), Gaussian seed (red: fast collapse, then a stall at $d \approx 0.02$ – 0.03 for ≥ 8 levels), and the real systems.

one-number readout of that shape. That the *fluctuations* of finite-time exponents, far more than their mean, carry this dynamical information has a clear precedent: Tomsovic and Lakshminarayanan showed that the variance and higher cumulants of the finite-time stability exponent in the standard map exhibit anomalous time-scalings that detect regular islands as small as 0.01% of phase space [29]. Our approach is complementary but distinct in three respects. We act on the *entire distribution shape* through an exactly-solvable renormalization operator rather than on low-order cumulant scalings. We obtain a *discrete two-class split* across five distinct systems, with the exponential distribution as an exact separatrix. And we trace the reported “ $\alpha^* \approx 4$ ” constant to a metastable plateau of that operator.

Relation to extreme-value theory. The dichotomy echoes, but is not reducible to, the Fisher–Tippett–Gnedenko trichotomy [17, 18, 21]: the exponential is simultaneously the Gumbel-domain excess law (Pickands–Balkema–de Haan [19, 20]) and the unique fixed point of pair-differencing (Puri–Rubin [14]). The cascade replaces threshold excess by iterated differencing, a renormalization semigroup in the spirit of [22] whose fixed-point structure appears to be new; it is the slow manifolds, not the fixed point itself, that carry the physics observed at finite depth.

Predictions — two now tested. The two-class picture makes five falsifiable predictions; two are tested here. (1) *Horizon dependence* — TESTED (§7): the outcome is stronger than predicted. The pendulum ensemble does not merely resist self-averaging; its FTLE variance is frozen outright ($\beta \approx 0$), because the release ensemble is a mixture over invariant structures. The class signature (above-separatrix reading, positive matched-null excess) holds at all ten horizons. (2) *Window signatures* remain open: non-mixed readings track the exponential f -curve (4) (verified here); mixed-phase readings should cross it only at markedly smaller f , when the sticky subpopulations leave the window. (3) *Out-of-sample systems* — TESTED (§8): Hénon–Heiles flips its matched-null excess positive exactly at the thin-web $\rightarrow$ mixed-sea transition and re-enters the self-averaging class as the sea dominates; one boundary regime (the escape energy) sits outside the two-class taxonomy and is flagged open. (4) *Deeper cascades* remain open: at $n \gg 10^7$ mixed-phase readings should continue the algebraic drift toward 3.3494 measured in §2.7–6; the class *contrast* at matched depth should persist. (5) A *microcanoni-*

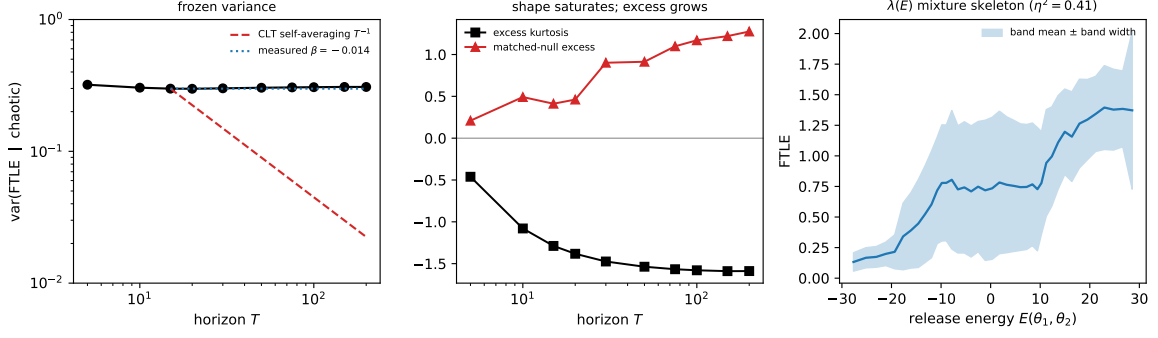


Figure 7: The ensemble structure of the pendulum constants. Left: chaotic-subset FTLE variance across a $13\times$ range of horizons — flat ($\beta = -0.014$), against the CLT’s T^{-1} . Middle: the shape saturates (excess kurtosis $\rightarrow -1.59$) while the matched-null excess grows. Right: the $\lambda(E)$ mixture skeleton over release energy (band mean $\pm$ band width), explaining $\eta^2 = 0.41$ of the variance; the bands themselves span the two-class taxonomy.

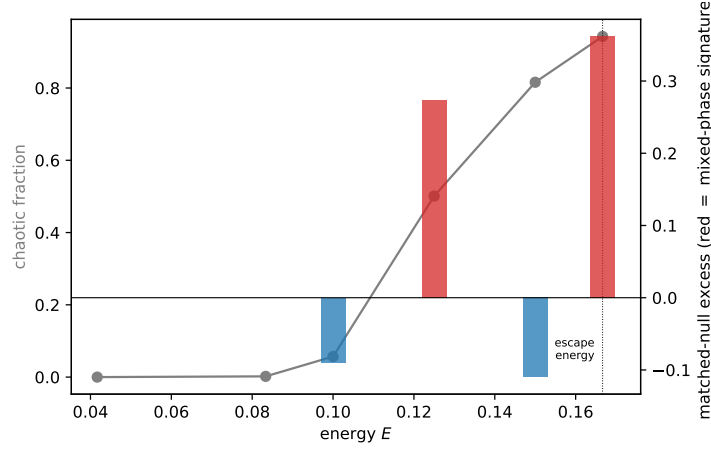


Figure 8: Hénon–Heiles across its transition: chaotic fraction (grey) and matched-null excess (bars; red = positive = mixed-phase signature) versus energy. The excess flips positive exactly where the mixed sea floods phase space ($E = 0.125$), returns negative as chaos dominates ($E = 0.150$), and flips again at the escape energy — the flagged boundary regime.

cal pendulum ensemble (single energy shell) at an energy where the shell is nearly uniformly chaotic should lose the platykurtic signature and read like the non-mixed class — the class membership of an ensemble is set by its ergodic architecture, not by the system’s name.

Open mathematical problems. (i) Spectral analysis of T linearized at Exp: our numerics indicate a nearly-degenerate contraction (d_k shrinking $\sim 5\%$ /level along the Gaussian manifold), and it remains to identify the slow eigenfunction and its eigenvalue. (ii) Classification of fixed shapes of $|A - B| \stackrel{d}{=} cX$ for $c < 1$ (the $c = 1$ case is [14]). (iii) A quantitative map from island-hierarchy exponents (e.g. the universal recurrence decay $z \approx 1.57$ [10]) to the platykurtic FTLE shape and its cascade excess. (iv) The empirical decay constant $r \approx 0.46\text{--}0.51$ of the mixed-phase excess: were it derived, it would make the plateau value itself computable.

10 A number-theoretic audit of the constants

We swept every constant in this study against a curated library of 67 named mathematical constants (transcendentals and their simple combinations; the Euler–Mascheroni, golden-ratio, Apéry, Catalan, Khinchin and Glaisher–Kinkelin constants; Riemann zeta values; the Feigenbaum and random-matrix-theory families; and KAM/standard-map constants), 159 low-order rationals m/n ($m, n \leq 16$) and the integers up to 1024, together with integer-relation (PSLQ) tests (`mpmath`, maximum coefficient 10^9 , 30-digit working precision) on the reported constant triplets, at match tiers $\tau_1/\tau_2/\tau_3 = 0.3\%/0.03\%/0.001\%$. We report the full audit so that no numerological reading survives by omission.

- $\eta_{0.2}(\text{Exp}) = 3.3494450 \dots$ has the exact form $1 + [5E_1(\ln 5)]^{-1}$ (eq. 4); the near-miss $10/3$ (0.48%, outside τ_1) is superseded by the closed form. Likewise $\eta_{0.2}(\text{HN}) = 4.4448$ and $\eta_{0.2}(\text{U}) = 9.6417$ are integrals, not numerology.
- The historical conjecture $\alpha^* = 4$ exactly is *excluded*: the flat-plateau parameter scan pins $3.896 [3.850, 3.942]$ and the 5×10^6 cascade gives $3.710 [3.675, 3.895]$. The speculative closed form $4 - 1/(4\pi^2) = 3.9747$ (motivated by a legacy pipeline value 3.9762 , which it matches to 0.04%) is likewise excluded by both modern intervals; under §6 such n -indexed plateau values are not expected to admit closed forms at all.
- Recurring loose coincidences are logged and rejected: class-mean/4 = $0.974 \approx K_c = 0.9716$ (0.25%, τ_1 ; no mechanism, two unrelated scales); noise-floor ratios $0.9241 \approx 1/\zeta(4)$ (0.02%, τ_2 ; a ratio of two near-equal noise scales); 3/16-like hits on standard errors. PSLQ finds no low-order integer relations among {class means, 4, noise scales} beyond spurious large-coefficient ones.
- The level-zero “constant” $\alpha_0 \approx 8\text{--}8.4$ across double-pendulum variants is explained to first order by (5) with the measured platykurtic bulks; its stability across variants reflects the stability of the *shape*, i.e. of the class, not an independent number — and it is horizon-indexed outright ($5.6 \rightarrow 14.9$ across $T = 5\text{--}200$, §7).
- The source program’s decay constant $r \approx 0.46$ “nearly constant across all systems” — and the derived quantity $\rho_2 = \log_2 r \approx -1.12$ on which a claimed mechanistic derivation was built — is the cumulant-halving transient of Proposition 4: $\log_2 r = -1$ exactly at leading order, with computable corrections. No free constant remains.
- A separate near-coincidence from the source program’s natural-unit runs — mean chaotic FTLE $0.398821 \approx 1/\sqrt{2\pi} = 0.398942$ (relative 3×10^{-4} at $n = 5 \times 10^6$, and seed-stable to 6×10^{-5} on one integrator arm) — fails constant-hood on the same grounds as everything else in this audit: a second integrator arm shifts it by 0.6%, i.e. it is protocol-indexed. Under §7 such ensemble means are mixture statistics; agreement with a named constant at one protocol point is numerology unless it survives protocol variation, which this does not.
- The relaxation exponent $\theta \approx 0.5\text{--}0.6$ of §2.7 is a measured quantity with a conjectured origin (marginal polynomial tower); deriving it exactly is an open problem, and we explicitly do *not* assign it a closed form.

11 Conclusions

A renormalization cascade that started life as an exploratory data-analysis device turns out to have exact structure underneath it: a unique exponential fixed point with a derivable, window-indexed reading (3.3494 at $f = 0.2$); a one-step universal collapse of all Gaussian bulks; an

exact cumulant flow under symmetrization (odd cumulants annihilated, standardized even cumulants scaled by 2^{1-m}) that derives the source program’s “universal” decay constant $r \approx 1/2$; a closed-form linearized eigenfamily $\mu(s) = 2/(2-s)$ sitting atop a unit-diagonal polynomial tower, a gapless, non-normal structure whose measured consequence is algebraic relaxation and metastable plateaus; and provable asymptotic tail conservation that finite windows never actually see. Recast on this foundation, a large-scale computational claim of a new universal constant “ $\alpha^* \approx 4$ ” of double-pendulum chaos resolves into something both humbler and more interesting than advertised. The constant turns out to be a plateau reading of a marginal flow, taken on an ensemble whose dispersion never self-averages ($\beta \approx 0$ across a $13\times$ range of horizons) because the canonical protocol builds a mixture, over energy shells, and within shells over coexisting invariant structures, whose bands internally recapitulate the very taxonomy found across systems. The universality that survives lives one level up: mixed-phase ensembles, identifiable by broad, flat-topped, platykurtic FTLE bulks, read strictly above the exponential separatrix, while fully chaotic, dissipative and near-regular ensembles, narrow or skewed and leptokurtic, read below it. This holds in eleven of eleven archetypal ensembles across five systems, survives a same-map/same-protocol control and a zero-within-class-heterogeneity meta-analysis, and holds up in an out-of-sample Hénon–Heiles sweep that flips sign on cue at the thin-web $\rightarrow$ mixed-sea transition (while honestly flagging one boundary regime, the escape energy, as beyond the taxonomy). The shape of the finite-time Lyapunov distribution, read through an operator with an exactly known reference point, turns out to be a practical probe of an ensemble’s ergodic architecture, and the exponential distribution, once again, sits at the center of the story.

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Data and code availability. The cascade operator, the calibration, linearization, ensemble and Hénon–Heiles analysis scripts, the constant-audit library, the raw FTLE ensembles (each with an SHA-256 manifest), and the result files that together reproduce every number and figure in this paper are archived at <https://github.com/RajveerKapoor/lyapunov-cascade-p1/tree/v1.0> (release v1.0); a file inventory is given in Appendix B.

A Derivations of the closed-form readings

Exponential. With $u = \ln(1/f)$ and $E \sim \text{Exp}(1)$ (memorylessness),

$$\mathbb{E}\left[\ln \frac{X}{u} \mid X > u\right] = \int_0^\infty e^{-t} \ln\left(1 + \frac{t}{u}\right) dt = \int_0^\infty \frac{e^{-t}}{u+t} dt \cdot [\text{by parts}] = e^u E_1(u),$$

giving (4). Values: $E_1(\ln 5) = 0.0851265$, so $\eta_{0.2} = 1 + (5 \times 0.0851265)^{-1} = 3.3494450$; similarly $\eta_{0.1} = 4.0873926$, $\eta_{0.3} = 2.9057918$, $\eta_{0.05} = 4.8111467$.

Half-normal (the universal Gaussian level-1). $u = \Phi^{-1}(0.9) = 1.281552$ (the 80th percentile of $|N(0,1)|$), and $\eta_{0.2} = 1 + \left[\frac{1}{0.2} \int_u^\infty \ln(x/u) 2\varphi(x) dx\right]^{-1} = 4.444794$ by numerical quadrature.

Uniform. $u = 0.8$: $\mathbb{E}[\ln(X/0.8) \mid X > 0.8] = 5 \int_{0.8}^1 \ln(x/0.8) dx = 0.115780$, so $\eta_{0.2} = 9.641716$. (Bounded supports read high: the log-excess is compressed.)

Gaussian level-zero law. For $X = |N(\mu, \sigma^2)|$, $cv = \sigma/\mu$ small: $u = \mu + \sigma z_f$ and $\mathbb{E}[X - u | X > u] = \sigma m_f$ with $m_f = \varphi(z_f)/f - z_f$; expanding $\ln(x/u) \approx (x-u)/u$ gives (5). Second-order corrections are $O(cv)$ relative and account for the observed 3–5% underprediction.

B Data provenance and reproducibility

- Double pendulum: `lyapunov_all_50k.npy` ($n = 5 \times 10^4$), `ftle_n500k_lagrangian_T15_wave0c.npy`, `ftle_n5M_lagrangian_T15.npy`; geometric variants `disc067_ftle_dp-{m0p5_l1, m2_l0p5, m0p5_l2}_n500k.T15.npy`.
- Standard map: `standard_map_K5_ftle_n500k.npy`; the $K = 1$ ensemble regenerated for this work ($n = 5 \times 10^5$, 10^3 iterations, seed 0xADD; its cascade fit 4.11/plateau 4.31 agrees with the program’s banked 4.382 [3.646, 4.934]).
- Lorenz-63: `lorenz_ftle_n500k_T100.npy`. Coupled quartic: `coupled_quartic_ftle_n500k.npy` ($\lambda = 0.1$), `coupled_quartic_lam0p2_ftle_n500k.npy`, with manifests recording integrator, seeds, moments, SHA-256.
- Meta-analytic quantities quoted in §5 (random-effects τ^2 , Cochran Q , cross- n correlation, ridge noise-null) were computed from the same production arrays; the corresponding analysis scripts accompany this manuscript as supplementary material.
- Synthetic calibration: 17 families, $n = 5 \times 10^5$ (deep runs 1.6×10^7), seeds 0xF1BE/0xADD/7; bootstrap-through-the-tree $B = 250$; full results in `synthetic_battery_results.json` and `addendum_results.json`.
- Horizon sweep (§7): `disc024_ftle_T{5..200}_N50000.npy` (canonical protocol re-runs at ten horizons, seed 42); release-angle grid `lyapunov_grid_100x100.npy` (legacy lineage; internal contrasts only). Analyses: `tsweep_analysis.py`, `energy_mixture.py` → `tsweep_results.json`, `energy_mixture_results.json`.
- Linearization (§2.7): `linearization.py` → `linearization_results.json` (cumulant identities on data, eigenfamily and polynomial-tower checks at 900 grid points, uniform-seed 8×10^6 relaxation).
- Hénon–Heiles (§8): `hh_sweep.py` → `hh_sweep_results.json` and `hh_ftle.E*.npy` ($n = 2.5 \times 10^5$ per energy, $T = 800$, Yoshida-4 variational Benettin, on-shell ICs to 6×10^{-17} , escape-guarded; seed 0x11E5).

C Full synthetic calibration table

D Estimator and extrapolation pathologies

Two failure modes of the published pipeline are documented so that readers do not mistake them for physics. (i) *Flat-sequence degeneracy*: the three-parameter geometric fit $\alpha_k = \alpha^* + Cr^k$ is unidentifiable when the sequence is flat (the exponential seed — its own fixed point): the optimizer runs to its bounds ($\alpha^* \rightarrow 12$, $r \rightarrow 0.99$) unless enough levels constrain it (≥ 14 levels at $n = 5 \times 10^6$ recover 3.3507). (ii) *Huge level-zero values*: when α_0 exceeds the fit’s initialization envelope (narrow bulks: Lorenz $\alpha_0 = 80$, $K=5$ $\alpha_0 = 53$, $|N(1, 0.1)|$ $\alpha_0 = 21$), the default fit fails or runs away; dropping level 0 restores a well-posed fit. The fit-free plateau statistic (median of late well-sampled levels) is immune to both and is reported throughout alongside fit values. Deep-level readings carry a small negative finite- n bias (even the exponential’s plateau reads 3.30–3.33 at levels with $n_k \lesssim 10^3$), common to all ensembles compared at matched depth.

Table 4: Synthetic battery at $n = 5 \times 10^5$, $f = 0.2$. Plateau = median of levels 2–6. Fit values marked ‡ ran to a parameter bound (Appendix D).

family	EVT domain	α_0	plateau	fit
Exp(1)	Gumbel	3.36	3.34	12.0‡
Exp(1) (5M)	Gumbel	3.35	3.35	3.351
half-normal	Gumbel	4.45	3.79	3.77
$ N(3, 1) $	Gumbel	8.64	3.89	3.87
$ N(1, 0.1) $	Gumbel	21.3	3.86	3.97*
lognormal $\sigma=0.5$	Gumbel	4.59	3.26	3.11
lognormal $\sigma=1$	Gumbel	2.79	2.70	7.9‡
Weibull $k=1.5$	Gumbel	4.52	3.68	1.6‡
Weibull $k=0.7$	Gumbel	2.64	3.01	3.33
Gamma(3)	Gumbel	4.77	3.58	3.58
uniform(0, 1)	Weibull	9.61	4.31	4.16
Pareto $a=2.5$	Fréchet	3.50	2.42	2.34
Pareto $a=4$	Fréchet	5.01	2.72	2.59
Pareto $a=6$	Fréchet	6.99	2.91	2.91
bulk + Pareto $a=3$ (12%)	Fréchet	3.10	2.66	2.27
bulk + Pareto $a=4$ (12%)	Fréchet	3.35	3.03	2.49
bulk + Pareto $a=5$ (12%)	Fréchet	3.53	3.28	2.27

*level-0 dropped before fitting.

Conflicts of Interest

The author declares no conflicts of interest.

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